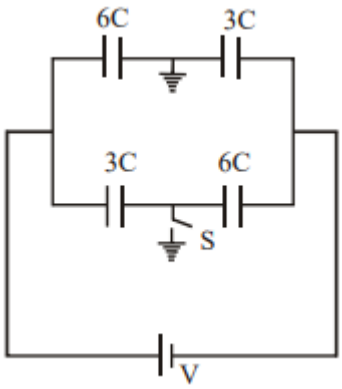
# Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

**Q.1**

Consider a network shown in figure. Initially the switch $S$ is open. The amount of charge flow through earth wire after closing the switch $S$ is -




**(A)** $CV$

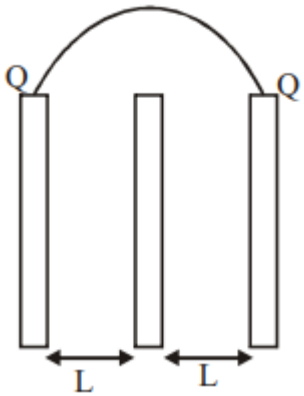
**(B)** $2CV$

**(C)** $\frac{1}{2}CV$

**(D)** $\frac{3}{2}CV$

**Q.2**

Three identical parallel large plates are placed at a distance $L$ as shown in figure. The outer plates are given charge $Q$ each and central plate is neutral. The outer plates are connected by a conducting wire. If the central plate is moved by distance $L/2$ left maintaining the parallelism, the charge flow through the connecting wire is -




**(A)** 0

**(B)** $Q/2$

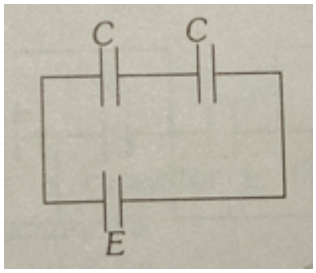
**(C)** $(2/3)Q$

**(D)** Q/3

**Q.3**

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Two identical parallel plate capacitor of capacitance $C$ each are connected in series with a battery of emf, $E$ as shown, if one of the capacitors is now filled with a dielectric of dielectric constant $k$, the amount of charge which will flow through the battery is (neglect internal resistance of the
battery)




**(A)** $\frac{k+1}{2(k-1)}CE$

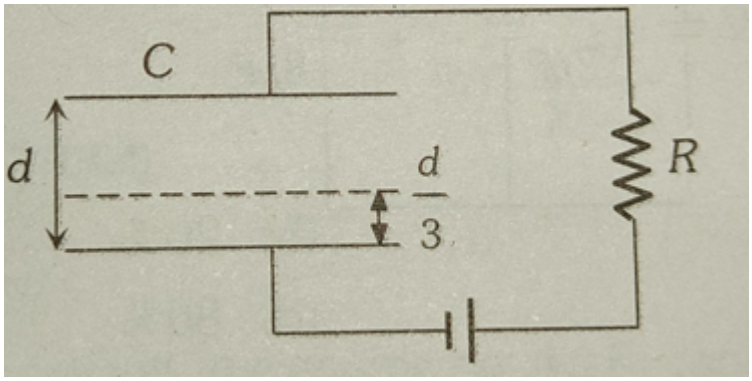
**(B)** $\frac{k-1}{2(k+1)}CE$

**(C)** $\frac{k-2}{(k+2)}CE$

**(D)** $\frac{k+2}{(k-2)}CE$

**Q.4**

A parallel plate capacitor $\mathrm{C}$ with plates of unit area and separation $d$ is filled with a liquid of dielectric constant $k = 2$. The level of liquid is $\frac{d}{3}$ initially. Suppose the liquid level decreases at a constant speed $\mathrm{V}$, the time constant as a function of time $t$ is




**(A)** $\frac{6\varepsilon_0 R}{5d+3Vt}$

**(B)** $\frac{(15d+9Vt)\varepsilon_0 R}{2d^2-3dVt-9V^2t^2}$

**(C)** $\frac{6\varepsilon_0 R}{5d-3Vt}$

**(D)** $\frac{(15d-9Vt)\varepsilon_0 R}{2d^2+3dVt-9V^2t^2}$

**Q.5**

A parallel plate capacitor of capacitance $C$ is connected to a battery and is charged to a potential difference $V$. Another capacitor of capacitance $2C$ is connected to another battery and is charged to potential difference $2\ \mathrm{V}$. The charging batteries are now disconnected and the capacitors are connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is

**(A)** Zero

**(B)** $\frac{25CV^2}{6}$

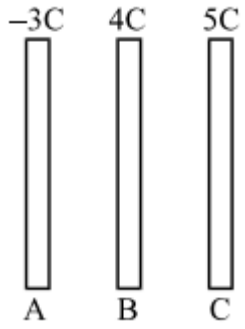
**(C)** $\frac{3CV^2}{2}$

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**(D)** $\frac{9CV^2}{2}$

**Q.6**

Three large metal plates $\mathrm{A}, \mathrm{B}$ and $\mathrm{C}$ are placed parallel to each other and charges are given as shown.




The charge on left surface of $B$, if $B$ and $C$ are earthed,

**(A)** $5\mathrm{C}$

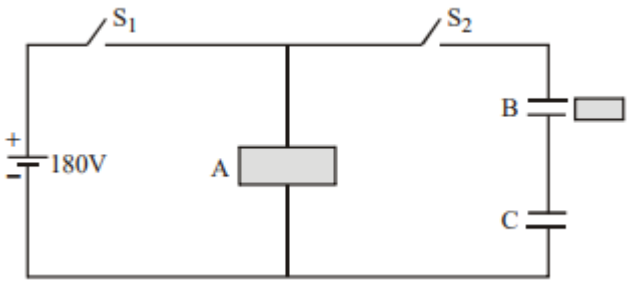
**(B)** $6\mathrm{C}$

**(C)** $3\mathrm{C}$

**(D)** $-3\mathrm{C}$

**Q.7**

In the circuit shown, capacitor A has capacitance $C_1 = 2\mu\mathrm{F}$ when filled with dielectric slab $(\mathrm{k} = 2)$. Capacitor $\mathrm{B}$ and $\mathrm{C}$ are air capacitors and have capacitances $\mathrm{C}_2 = 3\mu\mathrm{F}$ and $\mathrm{C}_3 = 6\mu\mathrm{F}$ respectively. Calculate the energy supplied by battery during process of charging when switch $S_1$ is closed alone




**(A)** $0.0324\ \mathrm{J}$

**(B)** $0.0648\ \mathrm{J}$

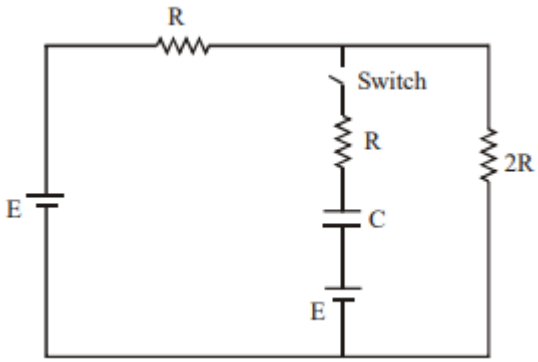
**(C)** $0.015\ \mathrm{J}$

**(D)** $0.030\ \mathrm{J}$

**Q.8**

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Initially the capacitor was uncharged. Current in the capacitor just after switching on will be -




**(A)** E/R

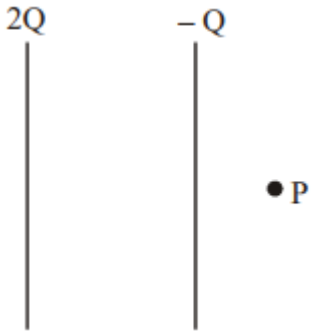
**(B)** E/2R

**(C)** E/5R

**(D)** E/4R

**Q.9**

In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is C. P is a point outside the capacitor and close to the plate of charge $-Q$. The distance between the plates is ' $d$ '.




**(A)** A point charge at point 'P' will experience electric force due to capacitor.

**(B)** The potential difference between the plates will be $3Q/2C$.

**(C)** The energy stored in the electric field on the region between the plates is $9Q^2/8C$.

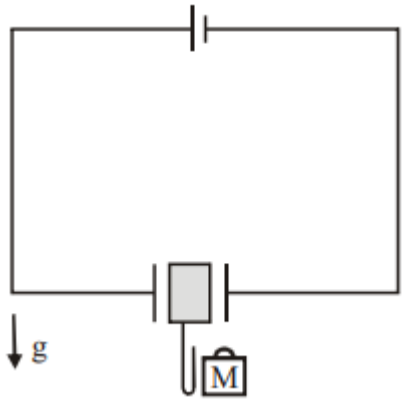
**(D)** The force on one plate due to the other plate is $\frac{Q^2}{2\pi\varepsilon_0 d^2}$.

**Q.10**

Figure shows a capacitor with its plates vertical and a hook is attached to a light dielectric slab and slab is initially at rest inside the plates of capacitor. Neglect the any frictional forces. Now a small block of mass $M$ is brought close to

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the U shaped hook and released suddenly to hang it in the hook -




**(A)** The dielectric with block may oscillate.

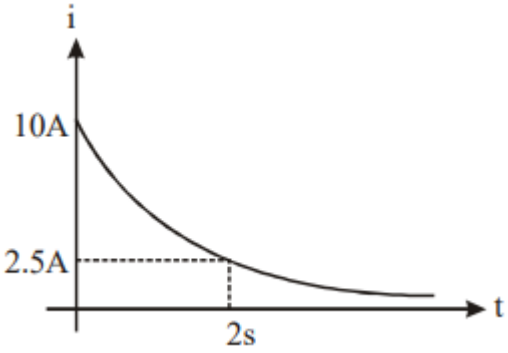
**(B)** The dielectric with block may be fall down by some distance and will come to permanently rest at equilibrium position.

**(C)** The dielectric with block may continue to fall and come out of the capacitor.

**(D)** The block may come to rest at the bottom of hook and dielectric is not displaced at all.

**Q.11**

The figure shows, a graph of the current in a discharging circuit of a capacitor through a resistor of resistance $10\Omega$.




**(A)** The initial potential difference across the capacitor is 100 volt.

**(B)** The capacitance of the capacitor is $\frac{1}{10\ln 2}$ F

**(C)** The total heat produced in the circuit will be $\frac{500}{\ln 2}$ joules.

**(D)** The thermal power in the resistor will decrease with a time constant $\frac{1}{2\ln 2}$ second.

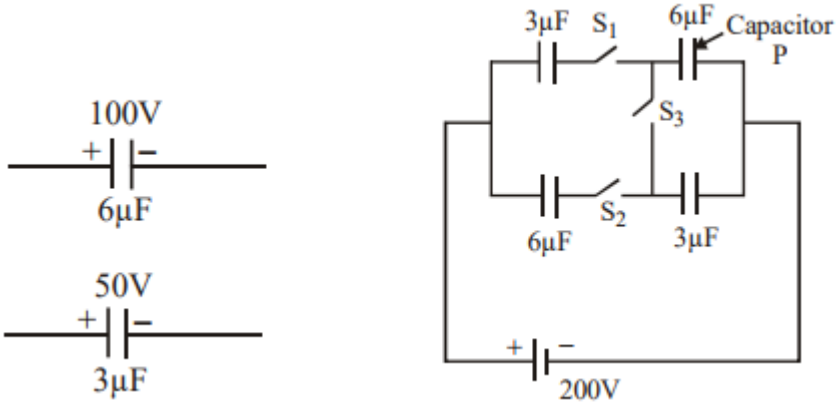
**Q.12**

**Passage**

A capacitor circuit consists of two $6\mu F$ and $3\mu F$ capacitors is initially charged to $100$ V and $50$ V respectively

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now connected to a source of emf $200\ \text{V}$ through the switches $\text{S}_1,\ \text{S}_2,\ \text{S}_3$ as shown in figure below:




**Question**

Charge on capacitor $P$ in steady state when $S_1$ is closed -

**(A)** $300\mu\text{C}$

**(B)** $700\mu\text{C}$

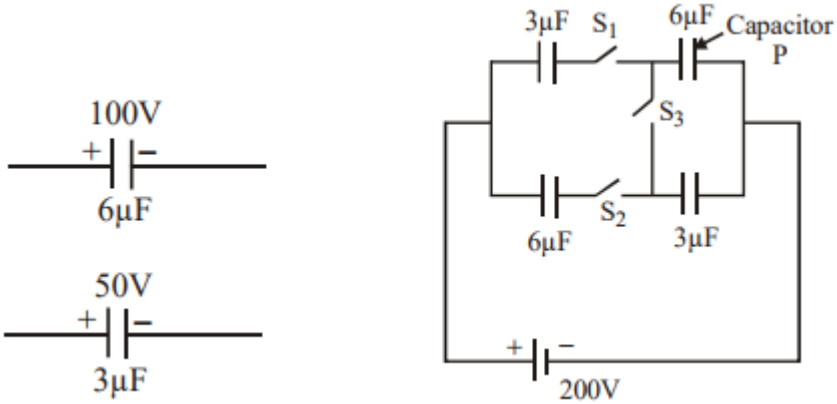
**(C)** $600\mu\text{C}$

**(D)** $100\mu\text{C}$

**Q.13**

**Passage**

A capacitor circuit consists of two $6\mu\text{F}$ and $3\mu\text{F}$ capacitors is initially charged to $100\ \text{V}$ and $50\ \text{V}$ respectively now connected to a source of emf $200\ \text{V}$ through the switches $\text{S}_1,\ \text{S}_2,\ \text{S}_3$ as shown in figure below:




**Question**

Now, switch $S_2$ is also connected, then charge on capacitor $P$ is -

**(A)** $300\mu\text{C}$

**(B)** $700\mu\text{C}$

**(C)** $600\mu\text{C}$

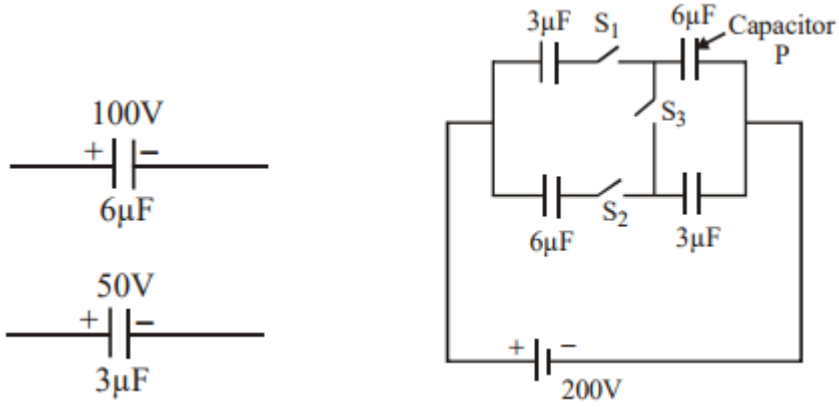
**(D)** $100\mu\text{C}$

**Q.14**

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**Passage**

A capacitor circuit consists of two $6\mu F$ and $3\mu F$ capacitors is initially charged to $100\ V$ and $50\ V$ respectively now connected to a source of emf $200\ V$ through the switches $S_1,\ S_2,\ S_3$ as shown in figure below:




**Question**

Now, switch $S_3$ is also connected, then charge on capacitor $P$ is equal to -

**(A)** $300\mu C$

**(B)** $600\mu C$

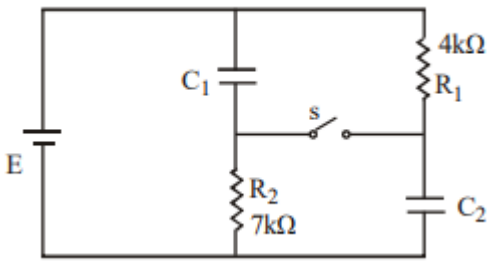
**(C)** $700\mu C$

**(D)** $100\mu C$

**Q.15**

**Passage**

The switch $s$ has been closed for long time and the electric circuit shown carries a steady current. Let $C_1 = 3.0\mu F, C_2 = 6.0\mu F, R_1 = 4.0k\Omega$, and $R_2 = 7.0k\Omega$. The power dissipated in $R_2$ is $2.8\ W$.




**Question**

The power dissipated to the resistor $R_1$ is -

**(A)** $2.8\ W$

**(B)** $1.6\ W$

**(C)** $4.9\ W$

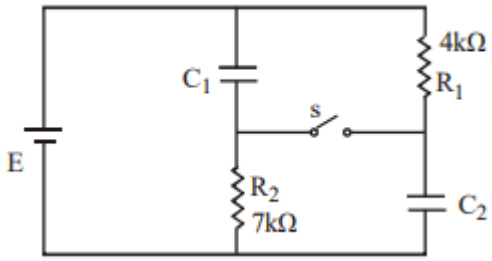
**(D)** $0$

**Q.16**

**Passage**

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The switch $s$ has been closed for long time and the electric circuit shown carries a steady current. Let $C_1 = 3.0\mu F, C_2 = 6.0\mu F, R_1 = 4.0k\Omega$, and $R_2 = 7.0k\Omega$. The power dissipated in $R_2$ is $2.8\ W$.




**Question**

The charge on capacitors $C_1$ and $C_2$ are respectively -

**(A)** $940\mu C, 940\mu C$

**(B)** $440\mu C, 440\mu C$

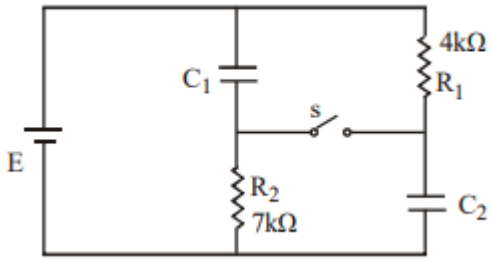
**(C)** $240\mu C, 840\mu C$

**(D)** $840\mu C, 240\mu C$

**Q.17**

**Passage**

The switch $s$ has been closed for long time and the electric circuit shown carries a steady current. Let $C_1 = 3.0\mu F, C_2 = 6.0\mu F, R_1 = 4.0k\Omega$, and $R_2 = 7.0k\Omega$. The power dissipated in $R_2$ is $2.8\ W$.




**Question**

Long time after switch is opened, the charge on $C_1$ is -

**(A)** zero

**(B)** $420\mu C$

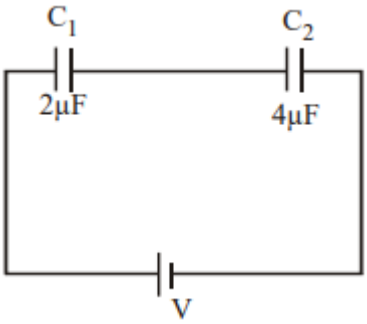
**(C)** $240\mu C$

**(D)** $660\mu C$

**Q.18**

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In the given figure, the separation between the plates of $C_1$ is slowly increased to double of its initial value then




| **Column I** | **Column II** |
|---|---|
| (A) the potential difference across $C_1$ | (p) increases |
| (B) the potential difference across $C_2$ | (q) decreases |
| (C) the energy stored in $C_1$ | (r) increases by a factor of 6/5 |
| (D) the energy stored in $C_2$ | (s) decreases by a factor of 18/25 |
| | (t) decreases by a factor of 9/25 |

**(A)** $(A)-s,t(B)-p,r,s,t(C)-q,s(D)-p,r,s,t.$

**(B)** $(A)-p,t(B)-p,r,t(C)-q,s(D)-p,r,s$

**(C)** $(A)-p,r(B)-q(C)-q,s(D)-q,t$

**(D)** $(A)-p,s,t(B)-p,r,s,t(C)-q,s(D)-t$

**Q.19**

One plate of a parallel capacitor $(5\mu\mathrm{F})$ has a fixed charge $10\mu\mathrm{C}$. The charge $\mathrm{q}(\text{ in } \mu\mathrm{C})$ on the other plate is varied with time ( $\mathrm{t}$ in seconds) as $\mathrm{q}=2\mathrm{t}$. The potential difference (in volts) between the plates will vary as $|\mathrm{x}-0.2\mathrm{t}|$. Find the value of $\mathrm{x}$.

**Q.20**

A particle of mass $10^{-3}\ \mathrm{kg}$ and charge $1.0\mathrm{C}$, is initially at rest. At time $t=0$, the particle comes under the influence of an electric field $\vec{E}(t)=E_0\sin\omega t\hat{i}$ where $E_0=1.0\mathrm{NC}^{-1}$ and $\omega=10^3\mathrm{rads}^{-1}$. Consider the effect of only the electrical force on the particle. Then the maximum speed, in $\mathrm{ms}^{-1}$, attained by the particle at subsequent times is

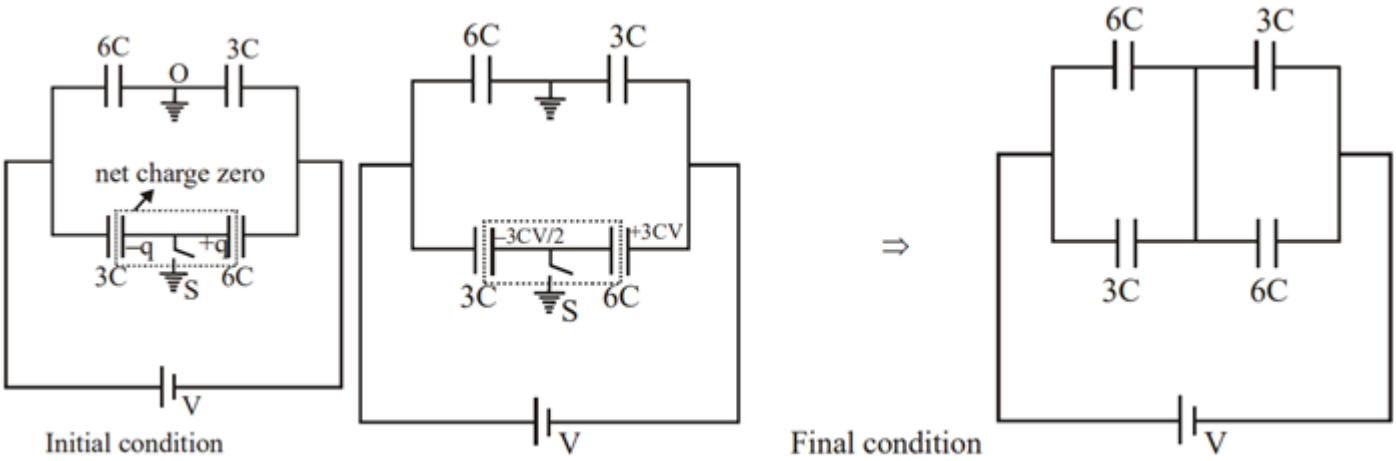
## Answers & Solutions

**Q.1 Answer:**

$\frac{3}{2}\mathrm{CV}$

**Solution:**

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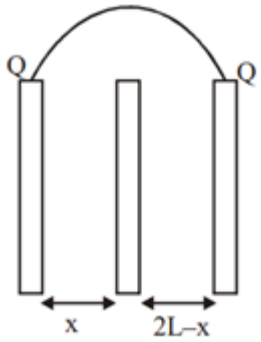



**Net charge in initial condition on the bounded system is zero. Final charge on the same system is** $+3\text{CV} - \frac{3\text{CV}}{2} = \frac{3\text{CV}}{2}$ **Therefore,** $\frac{3\text{CV}}{2}$ **charge will flow through the switch.**

**Q.2 Answer:**

0

**Solution:**




**The distribution of charge will not be effected by the motion of central plate.**

**The outer extreme faces of the left and right plates will have charges** $Q$ **each.**

**Q.3 Answer:**

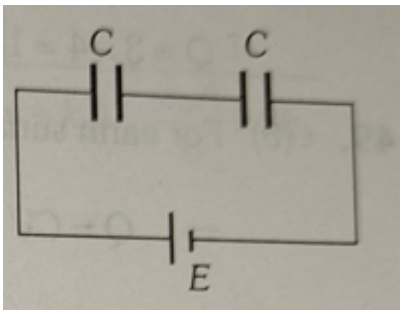
$\frac{k-1}{2(k+1)}CE$

**Solution:**

**Initial charge on both capacitor**

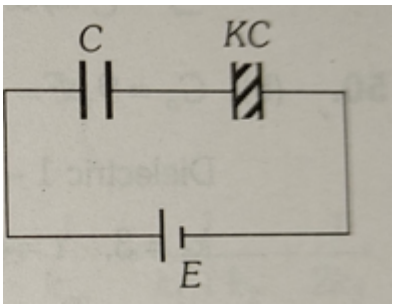
$$Q_i = \frac{CE}{2}$$

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**Now, charge on each capacitor**

$$Q_f = \left(\frac{KC}{K+1}\right) E$$




**Change in charge on $C$ is supplied by battery**

$$\begin{aligned}\text{Charge supply by battery } &= \left(\frac{KC}{K+1}\right) E - \frac{CE}{2} \\ &= CE\left[\frac{K-1}{2(K+1)}\right]\end{aligned}$$

**Q.4 Answer:**

$\frac{6\varepsilon_0 R}{5d+3Vt}$

**Solution:**

$A = 1$, **Separation** $= d$

$$k = 2, \quad t = \frac{d}{3}$$

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**Let level of liquid at any instant of time ' $t$ ' be $x$.**

$$V = -\frac{dx}{dt} \Rightarrow dx = -Vdt$$

$$\Rightarrow \quad \int_{\frac{d}{3}}^{x} dx = -V\int_{0}^{t} dt \Rightarrow x - \frac{d}{3} = -Vt$$

$$\Rightarrow x = \frac{d}{3} - Vt$$

**Now,** $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$

$$\Rightarrow \quad C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2} = \frac{\left(\frac{A\varepsilon_0}{d-x}\right) \times \left(\frac{\varepsilon_0 Ak}{x}\right)}{\frac{A\varepsilon_0}{(d-x)} + \frac{\varepsilon_0 Ak}{x}}$$

$$\Rightarrow \quad C_{eq} = \frac{kA\varepsilon_0}{kd + x(1-k)}$$

$$\Rightarrow \quad C_{eq} = \frac{2\varepsilon_0}{2d + \left(\frac{d}{3} - Vt\right)(1-2)} = \frac{2\varepsilon_0}{2d - \frac{d}{3} + Vt}$$

$$= \frac{6\varepsilon_0}{5d + 3Vt}$$

$$\text{Time constant } \tau = RC_{eq} = \frac{6\varepsilon_0 R}{5d + 3Vt}$$

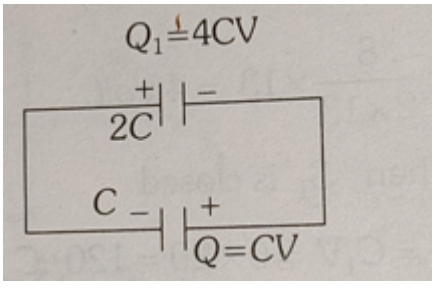
**Q.5 Answer:**

$\frac{3CV^2}{2}$

**Solution:**

$$\text{Total charge } (Q) = (2C)(2V) + C(-V) = 3CV$$

$$C_{eq} = 3C$$




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$$\text{Common potential } = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = \frac{Q}{C_1 + C_2}$$
$$= \frac{3CV}{3C} = V$$
$$U = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}(3C)V^2 = \frac{3}{2}CV^2$$

**Q.6 Answer:**

$6\mathrm{C}$

**Solution:**

**Charge induced on outer surface of plate $\mathrm{A}$ and $\mathrm{C}$ will be half of sum of the total charges.**

**So charge induced** $= \frac{-3+4+5}{2}\mathrm{C}$

$= \frac{6}{2}\mathrm{C}$

$= 3\mathrm{C}$

**Charge on soft side of plate $\mathrm{A}$ is $3\mathrm{C}$ so on right side of plate charge will be** $\Rightarrow 6\mathrm{C} - 3\mathrm{C} = 3\mathrm{C} + \mathrm{x}$ **(where $\mathrm{x}$ is charge on right)**

**To make total Charge on plate A-3C.**

**So charge on right side of plate A will be** $-6\mathrm{C}$ **and by law of induction charge on left side of plate** $B$ **will be** $+6\mathrm{C}$

**So,** $6\mathrm{C}$ **charge appears on left surface of plate** $\mathrm{B}$

**Q.7 Answer:**

$0.0648\ \mathrm{J}$

**Solution:**

$$\mathrm{q} = \mathrm{CV} = 2 \times 10^{-6} \times 180 = 360\mu\mathrm{C}$$

**Energy supplied by battery** $= \mathrm{qV} = 0.0648\ \mathrm{J}.$

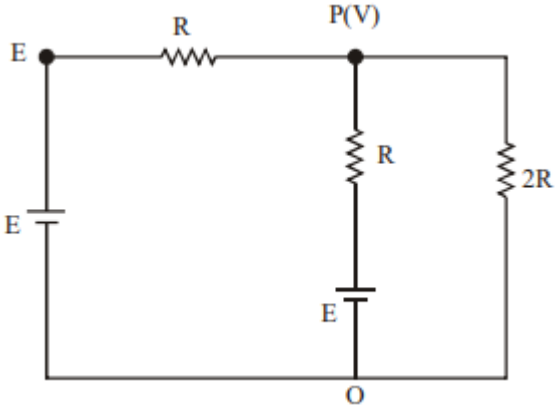
**Q.8 Answer:**

E/5R

**Solution:**

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**Just after switching, the capacitor will act like a conducting wire. So effective circuit will be**



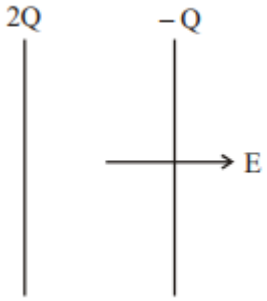

**By applying KCL at point, P,** $\frac{V-E}{R}+\frac{V-E}{R}+\frac{V}{2R}=0 \Rightarrow V=\frac{4E}{5}$

**So, current in the branch containing capacitor** $=\frac{E}{5R}$

**Q.9 Answer:**

A point charge at point 'P' will experience electric force due to capacitor.

**Solution:**




**(A) Due to net electric field P charge will experience force**

**(B)** $E=\frac{2Q}{2A\varepsilon_0}+\frac{Q}{2A\varepsilon_0} \Rightarrow E=\frac{3Q}{2A\varepsilon_0}$

$$E=\frac{3}{2}\frac{Q}{Cd} \Rightarrow Ed=\frac{3Q}{2C}=V$$

**(C) Energy** $=\frac{1}{2}\varepsilon_0 E^2 Ad=\frac{1}{2}\varepsilon_0\left(\frac{3Q}{2Cd}\right)^2 Ad=\frac{9}{8}\frac{Q^2}{C}$

**(D) Force on plate,** $F=\left(\frac{2Q}{2A\varepsilon_0}\right)\times\left(\frac{-Q}{1}\right)=-\frac{Q^2}{A\varepsilon_0}; F=\frac{Q^2}{A\varepsilon_0}$

**Q.10 Answer:**

The dielectric with block may oscillate.

**Solution:**

**If its velocity becomes zero before it comes out of the slab and force on the dielectric due to electric field upward is more than mg, it will oscillate, otherwise will fall out.**

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**Q.11 Answer:**

The initial potential difference across the capacitor is 100 volt.

**Solution:**

$V_0 = I_0 R = 10 \times 10 = 100$ **volts** **(since** $I_0 = 10$ **amp from figure)**

**Also,** $I = I_0 e^{-t/RC}$

$\ln\left(\frac{I_0}{I}\right) = \frac{t}{RC} \Rightarrow C = \frac{t}{R \ln(I_0/I)}$

**At** $t = 2\text{sec}, I = (2.5)A$

$C = \frac{2}{10 \ln(10/2.5)} = \frac{2}{10 \ln 4} = \frac{2}{10 \times 2 \ln 2} = \frac{1}{10 \ln 2}$

**Heat produced** $= \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{1}{10 \ln 2}\right)\left(100^2\right) = \frac{500}{\ln 2}$ **joules**

**Thermal power in resistance** $= i^2 R = \left(I_0^2 e^{-2t/RC}\right) R$

**Time constant for thermal power** $= \frac{RC}{2} = \frac{1}{2 \ln 2}$

**Q.12 Answer:**

$700\mu C$

**Solution:**

(i) $200 - \left(\frac{600+q}{6}\right) - \left(\frac{150+q}{3}\right) = 0$

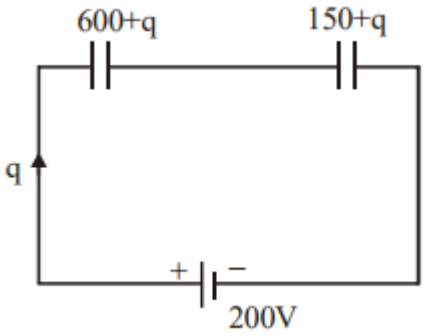
$1200 - 600 - q - 300 - 2q = 0$

$300 = 3q$

$q = 100\mu C$

**Charge on capacitor** $P = 600 + 100 = 700\mu C$

**Charge on** $3\mu F$ **capacitor** $= 250\mu C$




**Q.13 Answer:**

$700\mu C$

**Solution:**

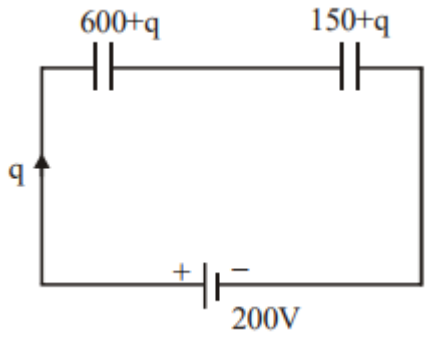
(i) $200 - \left(\frac{600+q}{6}\right) - \left(\frac{150+q}{3}\right) = 0\ 1200 - 600 - q - 300 - 2q = 0\ 300 = 3q$

$q = 100\mu C$

**Charge on capacitor** $P = 600 + 100 = 700\mu C$

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**Charge on $3\mu F$ capacitor $= 250\mu C$**




**(ii) Charge on capacitor P remains same.**

**Q.14 Answer:**

$600\mu C$

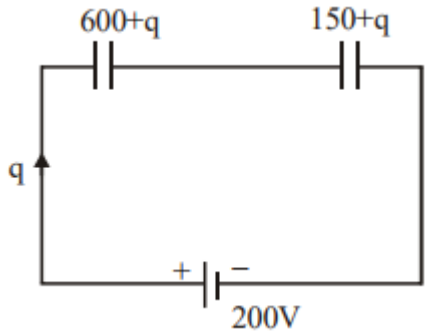
**Solution:**

**(i)** $200 - \left(\frac{600+q}{6}\right) - \left(\frac{150+q}{3}\right) = 0\ 1200 - 600 - q - 300 - 2q = 0\ 300 = 3q$

$q = 100\mu C$

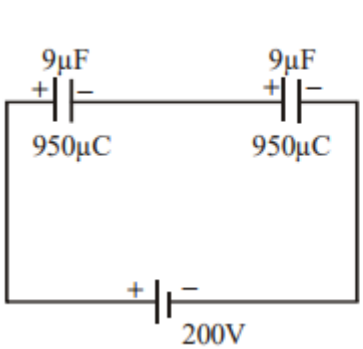
**Charge on capacitor** $P = 600 + 100 = 700\mu C$

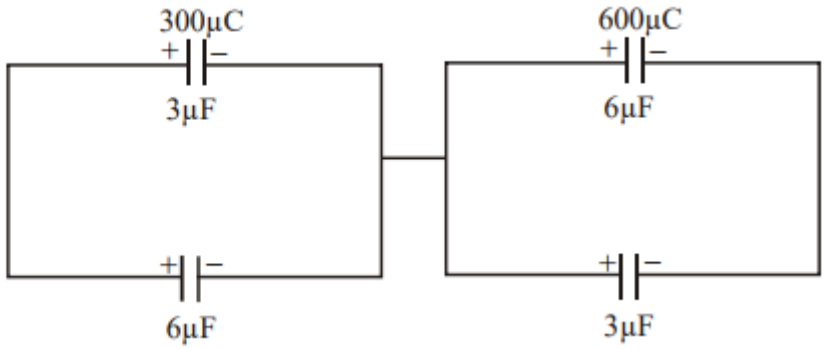
**Charge on $3\mu F$ capacitor $= 250\mu C$**




**(ii) Charge on capacitor P remains same.**

**(iii)** $200 - \left(\frac{950+q}{9}\right) \times 2 = 0$




$$900 - 950 - q = 0$$

$$q = -50\mu C$$

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**Charge on equivalent $9\mu F$**
**Capacitor $= 900\mu C$ then charge on $6\mu C = 600\mu C$**

**Q.15 Answer:**

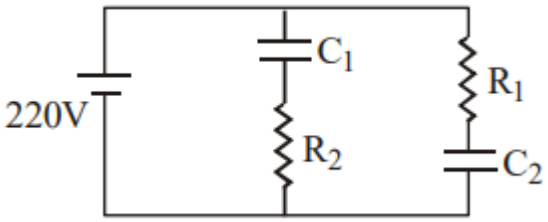
$1.6$ W

**Solution:**

$2.8 = i^2 \times 7 \times 10^3$

$i = 2 \times 10^{-2}$ A

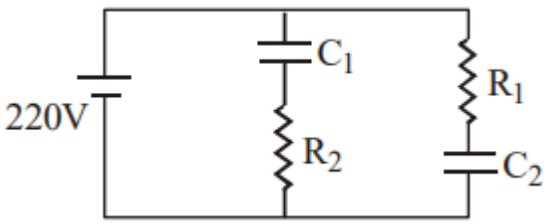
$P_{R_1} = i^2 R_1 = \left(2 \times 10^{-2}\right)^2 \times 4 \times 10^{-3} = 1.6$ W




**Q.16 Answer:**

$240\mu C, 840\mu C$

**Solution:**



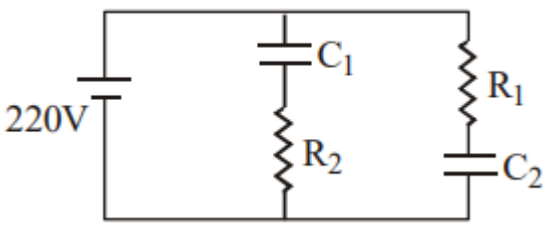

$Q_{C_1} = V_{R_1} \times C_1 = 80 \times 3 \times 10^{-6} = 240\mu C$

$$Q_{C_2} = V_{R_2} \times C_2 = 140 \times 6 \times 10^{-6} = 840\mu C$$

**Q.17 Answer:**

$660\mu C$

**Solution:**




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$$Q_{C_1} = EC_1 = 220 \times 3 \times 10^{-6} = 660\mu C$$

**Q.18 Answer:**

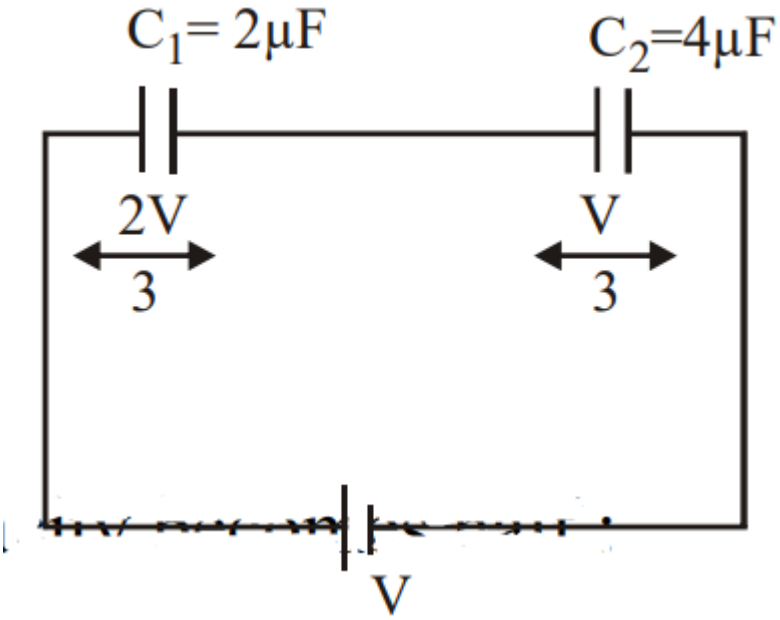
$(A) - p, r (B) - q (C) - q, s (D) - q, t$

**Solution:**

**(A) - pr, (B) - q, (C) - qs, (D) - qt**

$$U_i \text{ for } C_1 = C_1 \times \frac{4}{9} V^2 \times \frac{1}{2} = \frac{4\,V^2}{9}$$

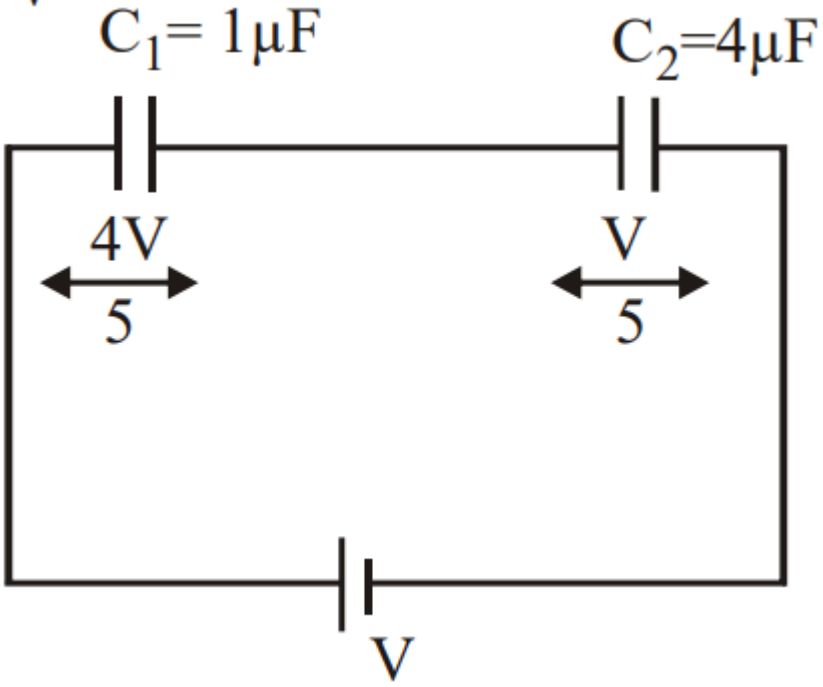
$U_i$ **for** $C_2 = 4 \times \frac{V^2}{9} \times \frac{1}{2} = \frac{2\,V^2}{9}$




**When separation of plates of $C_1$ gets doubled, its capacity becomes half:**

$$U_f \text{ for } C_1 = 1 \times \frac{16\,V^2}{25} \times \frac{1}{2} = \frac{8\,V^2}{25}$$
$$U_f \text{ for } C_2 = 4 \times \frac{V^2}{25} \times \frac{1}{2} = \frac{2\,V^2}{25}$$

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**Q.19 Answer:**

—

**Solution:**

**Let the charge on the other place be** $\mathrm{q}$

**Electric field between the plates** $= \left|\frac{\mathrm{q}-10}{2\varepsilon_0\ \mathrm{A}}\right|$**, where** $\mathrm{A}$ **= area of plate**

$\therefore$ **Potential difference between the plates** $= \left|\frac{\mathrm{q}-10}{2\varepsilon_0\ \mathrm{A}} \cdot \mathrm{d}\right|$

**where** $d$ **= distance between the plates**

$$= \left|\frac{q-10}{2C}\right| = \left|\frac{2t-10}{2(5)}\right| = |0.2t-1|$$

**Q.20 Answer:**

—

**Solution:**

$$E = qE_0 \sin \omega t = m\frac{dv}{dt}$$

$$\int_0^v dv = \frac{qE_0}{m}\int_0^{\pi/\omega} \sin \omega t dt$$

$$v = \frac{qE_0}{\omega m}(\cdot \mathrm{os}\ \omega t)_0^{\pi/\omega}$$

$$= -\frac{qE_0}{m\omega}((-\cos \pi) - (-\cos 0))$$

$$= \frac{2qE_0}{m\omega} = 2m/s$$

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